STATISTICS & PROBABILITY

MATH 2205 · Learning Summary

A visual, formula-focused study guide based on the UIU Summer 2026 course outline.

Coverage

- Introduction and types of data
- Frequency distributions and statistical graphs
- Measures of location and dispersion
- Moments, skewness and kurtosis
- Correlation and simple linear regression
- Probability rules and Bayes' theorem
- Discrete and continuous probability distributions
- Confidence intervals and hypothesis testing

Formulas are typeset as display equations, and graphs retain their natural proportions for readability.

1. Introduction

Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data to support decisions.

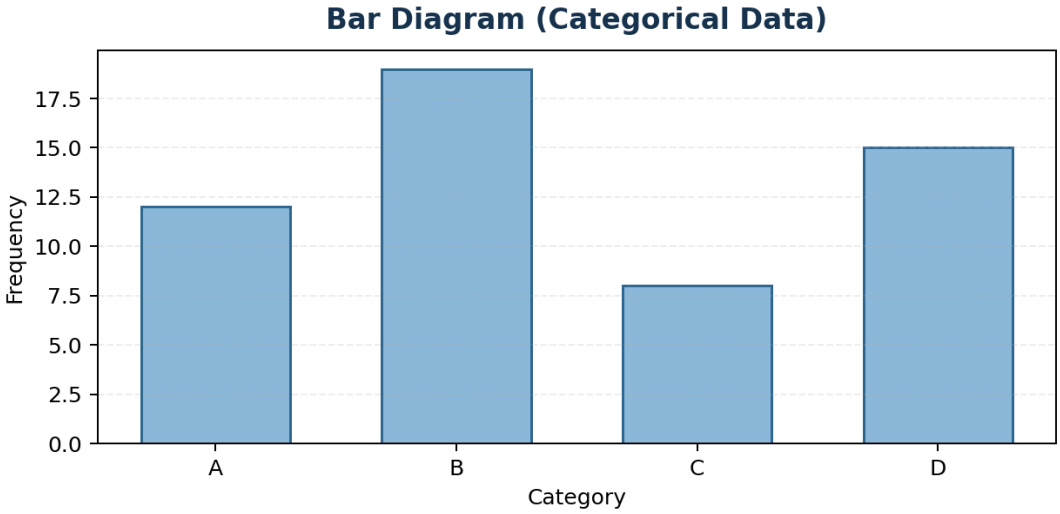
Term	Meaning
Population	The entire group about which we want information.
Sample	A subset selected from the population.
Parameter	A numerical description of a population (for example, μ or σ).
Statistic	A numerical description calculated from a sample (for example, $\bar{x}$ or s).
Variable	A characteristic measured for each individual or object.
Categorical data	Values represent labels or groups.
Quantitative data	Values represent counts or measurements; may be discrete or continuous.

Why sampling matters: a well-selected sample can provide useful information about a large population while reducing the time and cost of collecting data.

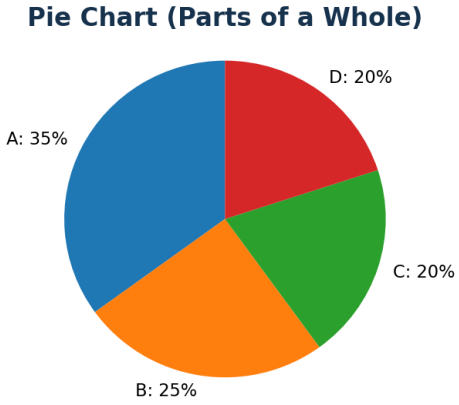
2. Summarizing Data

A frequency distribution organizes observations into categories or class intervals and records how often each occurs.

Common graphical displays

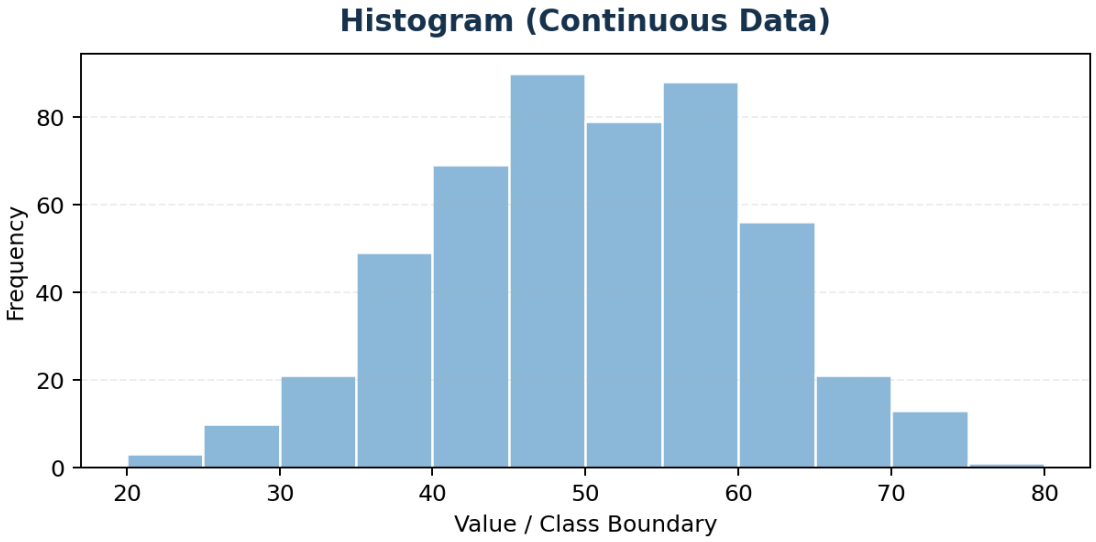


Use separated bars for categorical groups; bar height represents frequency or percentage.

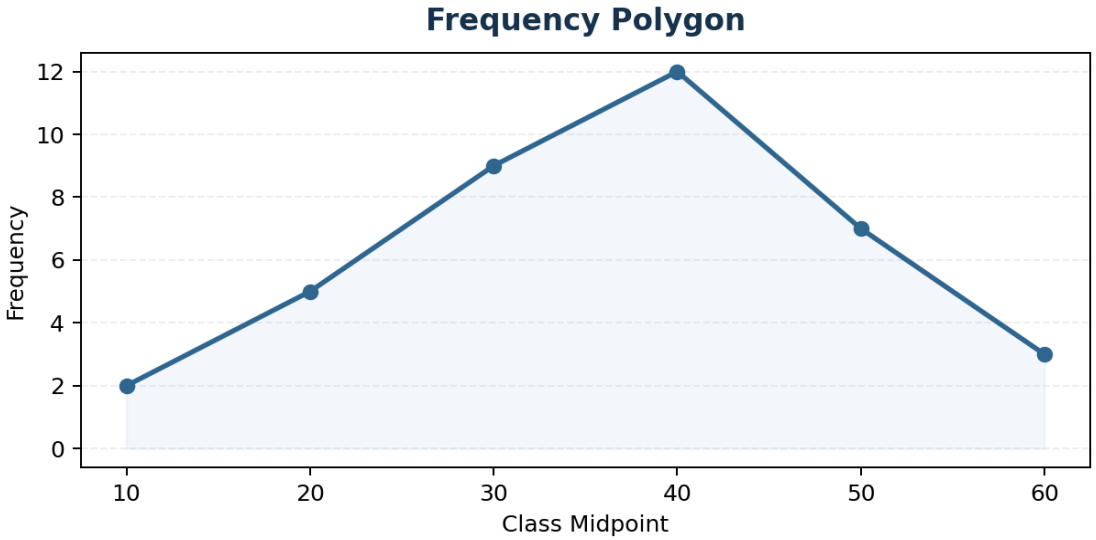


A pie chart shows each category as a share of the total. All slices together represent 100%.

2. Summarizing Data (continued)

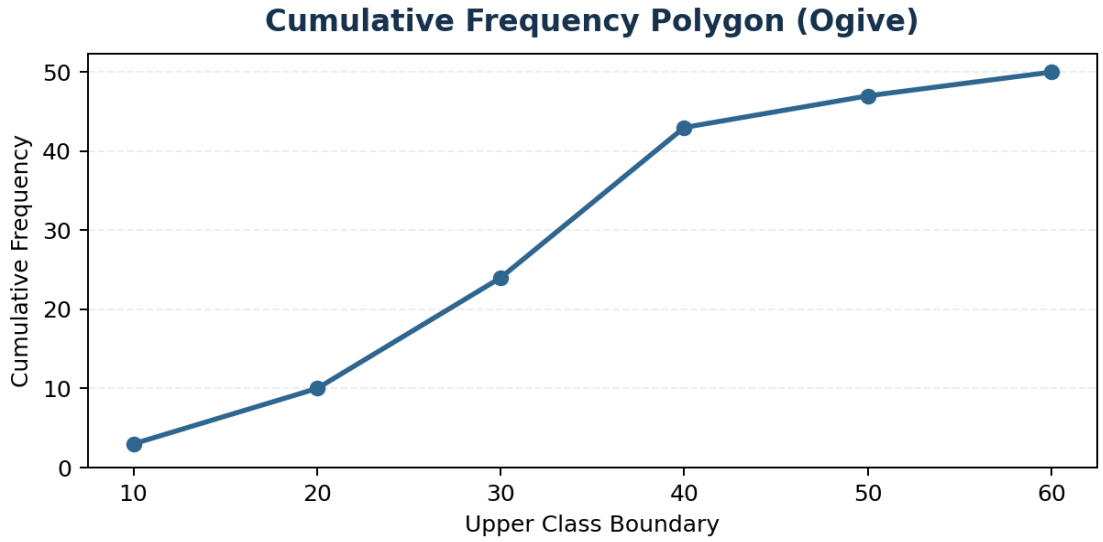


A histogram uses touching bars for numerical data grouped into class intervals.



A frequency polygon joins frequencies plotted at class midpoints.

2. Summarizing Data (continued)



An ogive plots cumulative frequency against class boundaries; it helps estimate medians, quartiles, and percentiles.

Stem-and-leaf diagram: split each observation into a stem (leading digit(s)) and a leaf (final digit). It preserves the original observations while showing the distribution.

Example

Data: 12, 14, 17, 21, 23, 25, 31 → Stem | Leaf
1 | 2 4 7
2 | 1 3 5
3 | 1
Key: 1 | 2 means 12.

3. Descriptive Statistics

Measures of location describe a typical or central value. Measures of dispersion describe how spread out the observations are.

Measures of location

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Sample arithmetic mean

$$GM = \left(\prod_{i=1}^{n} x_i \right)^{1/n}$$

Geometric mean (for positive observations)

$$HM = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Harmonic mean (for nonzero observations)

$$\text{IQR} = Q_3 - Q_1$$

Interquartile range: spread of the middle 50%

$$\text{Range} = x_{\max} - x_{\min}$$

Median: middle observation after sorting (for an even sample size, average the two middle values). Mode: most frequent value. Quartiles divide ordered data into four parts; deciles into ten; percentiles into one hundred.

3. Descriptive Statistics (continued)

Measures of dispersion

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

Sample variance

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

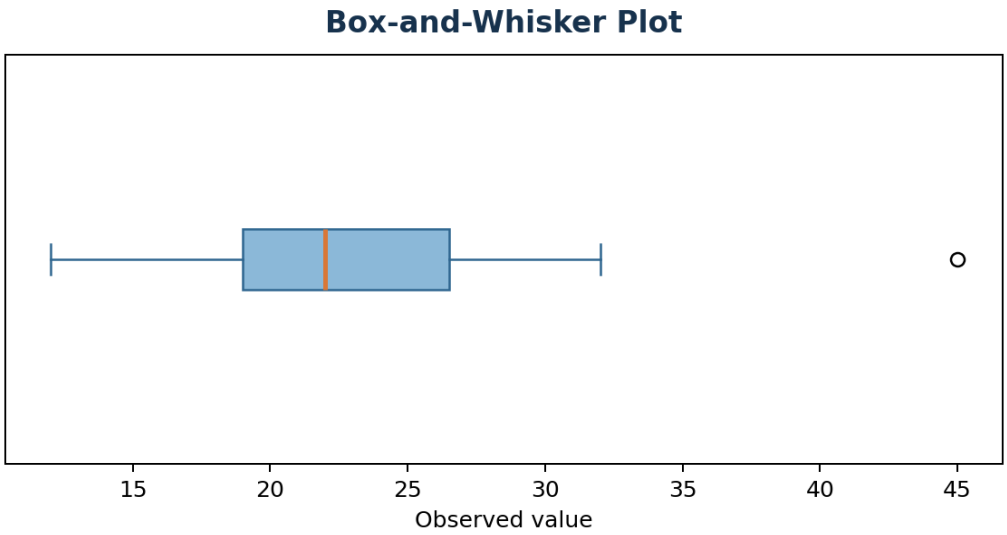
Sample standard deviation

$$MD = \frac{\sum_{i=1}^{n} |x_i - \tilde{x}|}{n}$$

Mean absolute deviation about the mean (one common definition)

$$CV = \frac{s}{\bar{x}} \times 100\%$$

Coefficient of variation; useful for comparing relative spread when means are positive



Five-number summary: minimum, Q1, median, Q3, maximum. The box spans Q1 to Q3; the line inside marks the median.

3. Descriptive Statistics: Moments, Skewness and Kurtosis

Moments summarize distribution shape. The second central moment is variance; higher moments describe features such as asymmetry and tail/peak behavior.

$$\mu_r = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^r$$

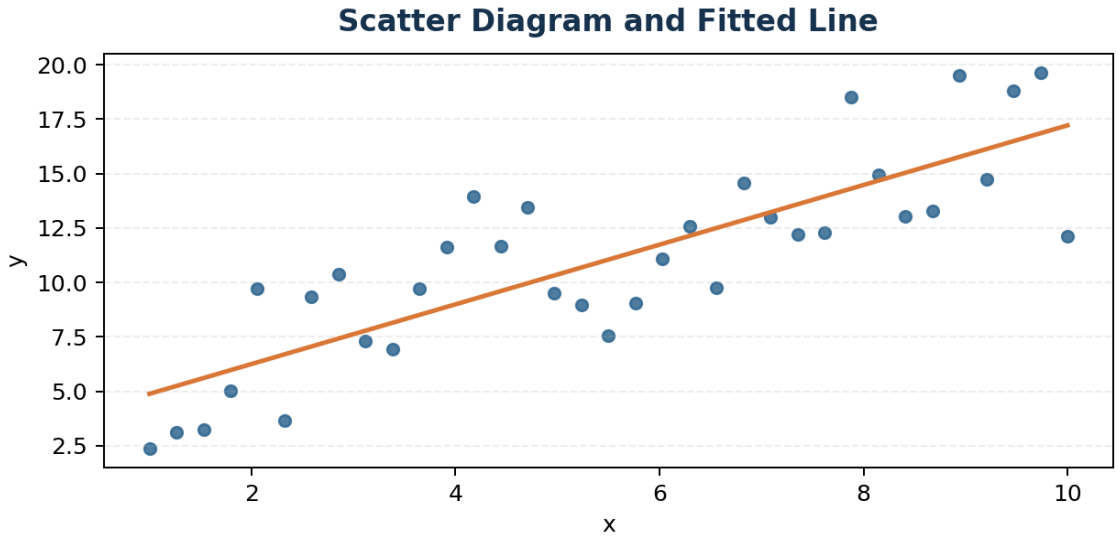
r-th population central moment

$$\text{Pearson skewness} = \frac{\mu_3}{\sigma^3}$$

Moment coefficient of skewness

Shape	Typical interpretation
Symmetric	Mean, median and mode are often equal or close.
Right-skewed (positive)	Longer right tail; mean is often greater than median.
Left-skewed (negative)	Longer left tail; mean is often less than median.
Kurtosis	Describes tail/peak behavior relative to a reference distribution; definitions may use raw kurtosis or excess kurtosis.

4. Simple Regression and Correlation



A scatter diagram displays paired observations. The fitted line summarizes the linear trend; correlation measures strength and direction of a linear relationship.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Pearson sample correlation coefficient, $-1 \le r \le 1$

$$\hat{y} = a + bx$$

Least-squares regression line

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \qquad a = \bar{y} - b\bar{x}$$

Slope and intercept

$$R^2 = r^2$$

For simple linear regression with an intercept, R^2 is the coefficient of determination.

Rank correlation uses ranks rather than raw values. For Spearman’s rank correlation with no tied ranks:

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

5. Probability

Probability ranges from 0 (impossible) to 1 (certain). The sample space S contains all possible outcomes; an event is a subset of S.

$$P(A^c) = 1 - P(A)$$

Complement rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

General addition rule

$$P(A \cap B) = P(A)P(B \mid A)$$

General multiplication rule

$$P(A \cap B) = P(A)P(B)$$

If A and B are independent

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Conditional probability

$$P(A_i \mid B) = \frac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)}$$

Bayes' theorem for mutually exclusive and exhaustive A_i

Concept	Meaning
Mutually exclusive	A and B cannot occur together: P(A ∩ B) = 0.
Independent	Knowing that one event occurred does not change the probability of the other.
Conditional probability	Probability of an event under the condition that another event occurred.

6. Probability Distributions

A random variable assigns a numerical value to each outcome. A probability distribution describes its possible values and their probabilities.

Discrete distributions

$$E(X) = \sum_{x} x\,P(X = x)$$

Expected value of a discrete random variable

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Variance

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Binomial distribution: n independent trials, success probability p

$$E(X) = np, \qquad \mathrm{Var}(X) = np(1-p)$$

Binomial mean and variance

$$P(X = x) = (1-p)^{x-1} p, \quad x = 1, 2, \ldots$$

Geometric distribution: trial number of the first success

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, \ldots$$

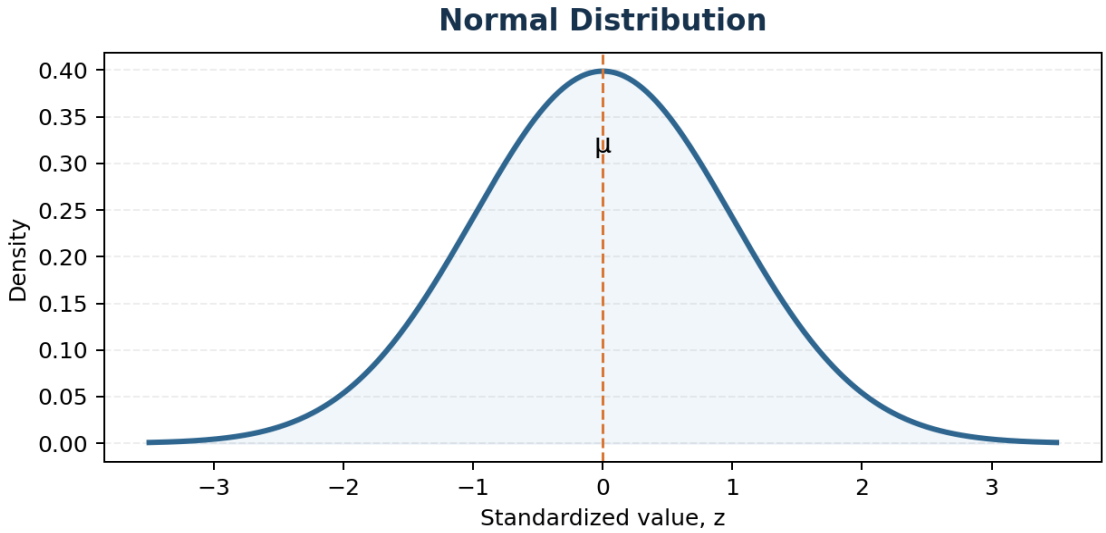
Poisson distribution: event count for a fixed interval

$$E(X) = \lambda, \qquad \mathrm{Var}(X) = \lambda$$

Poisson mean and variance

6. Probability Distributions (continued)

Continuous distributions



The normal curve is symmetric around μ. The total area under a probability density curve is 1.

$$Z = \frac{X - \mu}{\sigma}$$

Standard score for a normal random variable

$$X \sim N(\mu, \sigma^2)$$

Normal distribution notation

For a continuous random variable, probability is area under the density curve over an interval; P(X = a) = 0 for a single exact value. For independent random variables, expectations add; variances add when the variables are independent.

$$E(aX + bY) = aE(X) + bE(Y)$$

Linearity of expectation

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$$

When X and Y are independent

7. Statistical Inference: Confidence Intervals

Point estimation uses a sample statistic to estimate a population parameter. An interval estimate gives a range of plausible parameter values, with a stated confidence level.

$$\text{estimate} \pm (\text{critical value})(\text{standard error})$$

General confidence-interval structure

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Confidence interval for a population mean when σ is known (normal conditions apply)

$$\bar{x} \pm t_{\alpha/2,\, n-1}\frac{s}{\sqrt{n}}$$

Common interval for a population mean when σ is unknown, using Student's t distribution

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Large-sample confidence interval for a population proportion, when conditions are met

Confidence level	Common two-sided z critical value
90%	1.645
95%	1.960
99%	2.576

Interpretation: in repeated sampling, the method captures the true population parameter at the stated long-run rate. It is not correct to say that a fixed computed interval has that probability of containing the parameter.

8. Statistical Inference: Hypothesis Testing

A significance test assesses whether sample evidence is inconsistent with a null hypothesis H_0, using a preselected significance level α.

- State H_0 and the alternative hypothesis H_1 (or H_a).
- Choose the significance level α and an appropriate test statistic.
- Check the assumptions/conditions for the test.
- Calculate the test statistic and p-value, or compare with a critical value.
- Reject H_0 when p-value $\leq \alpha$; otherwise, fail to reject H_0.
- Write a conclusion in the context of the problem.

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

Common one-sample z statistic for a proportion under H_0

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

One-sample z statistic for a mean when population σ is known

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

One-sample t statistic for a mean when σ is unknown

Decision	Truth: H_0 true	Truth: H_0 false
Reject H_0	Type I error (probability α)	Correct decision
Fail to reject H_0	Correct decision	Type II error (probability β)

Failing to reject H_0 does not prove that H_0 is true. Statistical significance also does not automatically imply practical importance.

Quick Formula Reference

Topic	Key formula
Arithmetic mean	$\bar{x} = \Sigma x_i / n$
Sample variance	$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$
Standard deviation	$s = \sqrt{s^2}$
Interquartile range	$IQR = Q_3 - Q_1$
Pearson correlation	$r = \Sigma[(x_i - \bar{x})(y_i - \bar{y})] / \sqrt{\{\Sigma(x_i - \bar{x})^2 \, \Sigma(y_i - \bar{y})^2\}}$
Regression line	$\hat{y} = a + bx$
Conditional probability	$P(A\|B) = P(A \cap B) / P(B)$
Bayes' theorem	$P(A\|B) = P(B\|A)P(A) / P(B)$
Binomial probability	$P(X=x) = C(n,x)p^x(1-p)^{n-x}$
Poisson probability	$P(X=x) = e^{-\lambda} \lambda^x / x!$
Standard score	$z = (x - \mu) / \sigma$
Confidence interval	estimate ± critical value × standard error

Learning outcomes reflected in these notes: describe data with appropriate graphs and numerical summaries; interpret correlation and regression; apply probability rules and standard distributions; construct confidence intervals; and conduct basic significance tests.